MTH2008

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

January 2023

Real Analysis

Module Leader: Prof Ana Rodrigues

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Let $S = (-\infty, -1) \cup \{0\}$.

   (i) Is S open, closed, neither or both? Justify your answer.

   (ii) Find the interior S° and the complement S^c of S.

   (iii) Find the set of limit points of S.

   (iv) Find the boundary ∂S of S. (8)

   (b) Using the $\epsilon - \delta$ definition of a limit, prove that

$$\lim_{x \to 0} x \cos\left(\frac{1}{x^3}\right) = 0.$$

   (10)

   (c) Using the $\epsilon - \delta$ definitions of limits and continuity, prove that the function $f : [0, +\infty) \to \mathbb{R}$ defined by

$$f(x) = \sqrt{8x} - 4$$

   is continuous from the right at 2. (10)

   (d) Find $\lim_{n \to \infty} s_n$ for the following sequence, justifying your answer using the $\epsilon - N$ definition of a limit:

$$s_n = \frac{n}{n + \sqrt{n}}.$$

   (10)

   (e) Use Riemann sums to show that the function

$$f(x) = x^2 + 4x$$

   is integrable on $[0, 1]$ and hence determine the value of its definite integral. (12)

   [**50**]

SECTION B

2. (a) Let A be a non-empty subset of $\mathbb{R}$ bounded above. Set

$$B = \{-a |\ a \in A\}.$$

Show that B is bounded below and that

$$\inf B = -\sup A.$$

(10)

(b) Let $f : D \to \mathbb{R}$. Let $x_0 \in D$ be such that $\lim_{x \to x_0} f(x) = l$ exists. Using the $\epsilon - \delta$ definition of a limit prove that

$$\lim_{x \to x_0} |f(x)| = |l|.$$

(5)

(c) Using the $\epsilon - \delta$ definition of a limit prove that

$$\lim_{x \to \infty} \frac{1}{x^3 + 2x + 1} = 0.$$

(10)

[**25**]

3. (a) Let $[a, b]$ be a closed and bounded set in $\mathbb{R}$. Let $f : [a, b] \to \mathbb{R}$ be a continuous function, differentiable on (a, b).

(i) State the *Mean Value Theorem*.

(ii) Prove that $f'(x) = 0$ for all $x \in (a, b)$ if and only if f is constant on (a, b).

(10)

(b) Let $\{x_n\}$ be a sequence of real numbers such that the two subsequences $\{x_{2n}\}$ and $\{x_{2n+1}\}$ converge to the same limit. Is the sequence $\{x_n\}$ necessarily convergent? Justify your answer. (7)

(c) Consider the function $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(x) = \|x\|^2$ (where $\|x\|$ is the Euclidean norm of x). Is it true that f is differentiable for every $p \in \mathbb{R}^n$? (8)

[**25**]

4. (a) Let $f : [a, b] \to \mathbb{R}$ be a bounded function on the interval $[a, b]$, and let $P = (x_0, x_1, \ldots, x_n)$ be a partition of $[a, b]$. Give the definition of the lower and upper Riemann sum of f with respect to the partition P. (5)

(b) Prove, using an ϵ-δ argument, that the function $f : \mathbb{R}^2 \to \mathbb{R}$, given by

$$f(x) = \begin{cases} \dfrac{x_1^2 x_2^2}{x_1^2 + x_2^2} & x \neq 0, \\ 0 & x = 0, \end{cases}$$

is continuous at $x = 0$. (8)

(c) Consider the function $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f((x_1, x_2)^T) = (\cos(x_1 x_2), \sin(x_1 x_2))^T.$$

(i) Find the derivative of f.

(ii) Show that f is differentiable at $(x_1, x_2)^T = (\pi, 1)^T$. (12)

[25]